

# Unit 1: Fundamentals of UX Design

## CHAPTER I

### Chapter I Agenda:

1. What is UI & UX?
2. Difference between UI and UX.
3. Principles of Good Design.
4. What is Design Spirits?
5. Key Elements of UX.
6. Design for different Platforms
7. The UX Design Process: Research, Design, Prototyping, Testing, Implementation.

### What is UI & UX?

The user experience is how a person, the user, feels about interacting with or experiencing a product. A product is a good, service, or feature.

It might be a physical product, like a video game controller or a bag of potato chips, or a technology product, like an app, website, or smartwatch. For example, for a user to have a good experience, the product needs to be usable, equitable, enjoyable, and useful. User experience is about improving usability or making something easier to use. This means that the design, structure, and purpose of the product are clear to everyone.

Think about a ketchup bottle. Historically, ketchup came in a glass bottle. The user had to hit the bottom of the glass to make ketchup come out. Often, no ketchup or too much ketchup would come out of the bottle. Today, the ketchup bottle has been redesigned into a plastic squeezable bottle, which makes it easier to use and allows users to control how much ketchup comes out. UX designers need to think about every person who uses the product. This might include people with disabilities, or people with very different life experiences from your own.

For example, one user might find an app with a lot of text easy to use. On the other hand, a user with a visual impairment might want different features like sound. Considering the unique needs of many different people is important work and thinking about equitable design is key. Being equitable means your designs are useful and marketable to people with diverse

abilities and backgrounds. User experience is also about making things enjoyable to use, which creates a positive connection between the user and the product. UX designers foster that positive connection by taking a user's thoughts and feelings into account when making products. Think about ordering takeout food online.

When you search for a restaurant on Google or Yelp, you can see photos of dishes and read reviews from other people who have eaten there. This experience is enjoyable because you can make a more informed choice about what to order, leaving you feeling happier about the product. So, how do we figure out what makes a user happy with the product? That's where research comes in. To know how users feel, we have to collect evidence on how they're experiencing that product in real time and ask them about it, too. As humans, we want products that are useful, meaning they solve our problems.

For example, if you're lost, a map app telling you how to get home is useful. But if the app can't find your current location, it's not so useful anymore. So now we know what it means for a user to have a good experience. Why is it important to business? In 2018, the research firm McKinsey & Company studied companies in three industries: medical technology, consumer goods, and retail banking. They found that, regardless of industry, businesses that focused on good usability and design performed better than their competitors. Basically, it comes down to this: When people like a product, they use that product a lot, and they recommend it to their friends.

And more people using the product means better business for the company. Plus, when users have a good experience with the product, they're more likely to have a positive opinion of the company that made it. A win-win for the user and the business. Can you think of a product you loved so much that you shared it with a friend? If you did, that's a great example of user experience design in action. Despite all this, a lot of companies don't understand the important role that UX design can play in improving their business, and that's where you as a UX designer come in. So now you know how to describe user experience and UX design.

## The Difference Between UX and UI

One of the most common misconceptions in the world of design is the interchangeable use of the terms "UX" (User Experience) and "UI" (User Interface). While closely related, they refer to distinct aspects of the design process, each playing a crucial role in creating a successful product.

**User Interface (UI)** design focuses on the visual elements of a product—the layout, typography, color schemes, buttons, icons, and overall aesthetic that users interact with. It's about crafting the look and feel of a product, ensuring that each visual element is intuitive, appealing, and functional.

On the other hand, **User Experience (UX)** design is concerned with the overall experience a user has when interacting with a product or service. UX design encompasses everything from usability and accessibility to the emotions and satisfaction users derive from the interaction. It focuses on the flow, structure, and logic behind the interface, ensuring that users can achieve their goals efficiently and enjoyably.

In short, UI design is about how a product looks, while UX design is about how it works and feels. Think of UX as the blueprint and the foundation, guiding the overall structure of the experience, while UI adds the finishing touches that make the experience engaging and visually pleasing.

## Characteristics of good user experience

Good design is easy to spot but often hard to pin down. What exactly makes a product effective to its users? Is it a matter of simplicity, structure, or functionality? The answer depends on the product in question. Some primary characteristics of good UX —**usable**, **equitable**, **enjoyable**, and **useful**—can help you evaluate a product's design.

If a product is **usable**, it means the design, structure, and purpose of the product is clear and easy to use. As you evaluate a product for usability, you can ask questions like: Is everything in the design easy to find? Is the design's functionality easy to understand? Can users accomplish specific tasks within the design? As you evaluate, these questions can help you determine whether the design delivers a usable experience.

Imagine you are evaluating the usability of an airline app. Assuming the primary purpose of this app is to book a flight, the design should provide a clear and easy way to complete that task. For example, a section where you can easily enter travel and flight details on the homepage would be an example of good usability.

If a product is **equitable**, it means a design is helpful to people with diverse abilities and backgrounds. In other words, the product's design addresses the needs of a diverse audience and ensures a high-quality experience is delivered to all users regardless of background, gender, race, or ability. Equity means providing people with the tools they need to accomplish their goals and support improved quality of life. Equity goes beyond the concept of equality, where everyone is given equal resources, because people often need different tools and support based on their needs. This is especially important to keep in mind for those in commonly disenfranchised groups. As you evaluate the equity of a product's UX, you can ask questions like: Are the needs of a diverse group of users considered? Does the product's design address the needs of traditionally underrepresented and excluded groups? These questions can help you determine whether the design delivers an equitable experience.

Imagine you are evaluating how equitable a social messaging app is. You might consider the design more equitable if the keyboard emoji list includes different skin tones and gender-neutral avatar options.

If a product is **enjoyable**, it means the design delights the user. The design reflects what the user may be thinking or feeling and creates a positive connection with them. A product's design doesn't have to be enjoyable for it to function properly. But, an enjoyable design adds to an already functional product and can enhance the user's feelings about the experience. As you evaluate how enjoyable a product's UX is, you can ask questions like: Are there aspects of the design that consider the user's feelings? Does the design inspire delight in the user? Does the design keep the user engaged throughout their experience? These questions can help you determine whether the design delivers an enjoyable experience.

Imagine you are evaluating how enjoyable a video streaming app is. Design aspects that might increase how much you enjoy the product include personalized recommendations based on previous watching habits, or the ability to customize the appearance of your account.

If a product is **useful**, that means it solves user problems. In other words, the design intentionally solves a user problem that the designer has identified. It's important to note that, while similar, useful and usable have different meanings. A product that is useful isn't always usable. The same is true for the opposite. The distinction between the two is that usability refers to the product working well and being easy to use, while usefulness refers directly to the ability to solve user problems. As you evaluate how useful a product's UX is, you can ask questions like: Does the design add value to the user's experience? Does the design solve a problem for the user? Does the design help the user achieve a specific goal? These questions can help you determine whether the design delivers a useful experience.

Imagine you're evaluating how useful a banking app is. Users typically download these apps because they need a place to manage their money. With this in mind, aspects of the app that might be considered useful are features that can be used to transfer money between accounts and pay bills.

# Characteristics of a Good User Experience

## Useful – does it solve a real problem?

It solves a real problem: getting from Point A to Point B without getting lost. If it didn't do this reliably, nothing else about it would matter.

## Usable – can people actually do the task?

You type a destination and instantly get turn-by-turn directions – no manual, no tutorial needed. Even a first-time user can operate it within seconds.

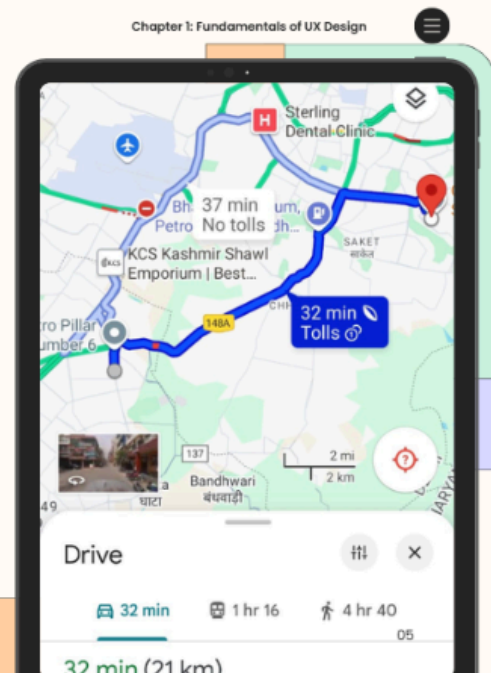
## Equitable – does it work for people with different abilities/backgrounds?

It adapts to different users' needs: voice navigation for people who can't look at the screen while driving, walking/wheelchair-accessible route options, offline maps for low-connectivity areas, and multi-language support for non-English speakers.

## Enjoyable – does it create a positive feeling?

Small emotional touches make it feel good, not just functional – live traffic colors that reassure you, the reassuring recalculating voice when you take a wrong turn instead of scolding you, ETA updates that reduce anxiety.

By Premitha Kamath



## Principles of Good UX Design:

Good UX design doesn't happen by accident. It follows core principles that guide designers in creating products that are not only functional but also enjoyable and valuable for users.

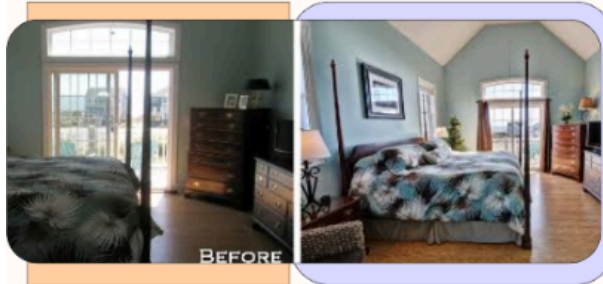
Below are some of the fundamental principles of good UX design:

1. **User-Centered Design:** The cornerstone of UX is designing with the user in mind.

This means conducting thorough research to understand the users' needs, behaviors, and pain points, and designing solutions that address them directly.

User-centered design ensures that every decision made during the design process is grounded in the user's perspective.

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**Remember: "You are not the user."**

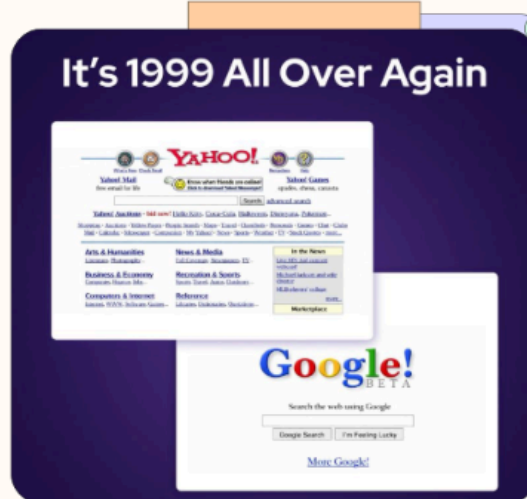
Airbnb almost failed in its early years. The founders assumed the problem was pricing and marketing. It wasn't until they flew to New York, met hosts in person, and saw their apartments and listing photos firsthand that they realized the real problem: the photos were bad, so nobody trusted the listings. They brought in professional photography — bookings doubled within a week. The fix wasn't a feature. It was actually seeing the user's world.

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2. **Simplicity and Clarity:** A simple design is often the most effective. Clear, concise interfaces reduce the cognitive load on users, making it easier for them to navigate and interact with the product. Unnecessary complexity can lead to confusion and frustration, so prioritizing simplicity is key to a positive user experience.

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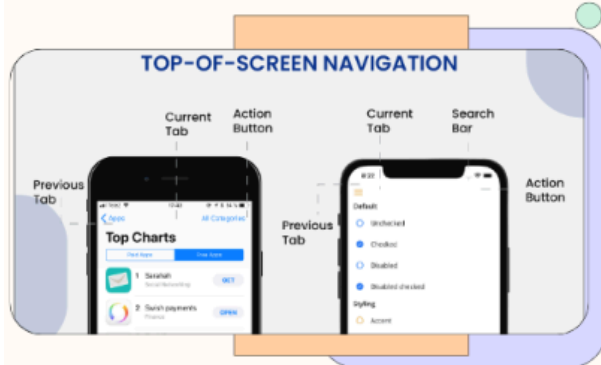
**Less thinking required = better experience.**

Google's homepage. One logo, one search box, one button. Compare that to early-2000s portals like Yahoo or MSN, which crammed news, ads, weather, and stock tickers onto the homepage. Google won not by doing more, but by doing almost nothing — and getting you to your goal in one action.

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3. **Consistency:** Consistency in design helps users build familiarity with the product. This includes maintaining uniformity in layout, visual elements, language, and interactions across the product. Consistent design creates a sense of predictability, allowing users to feel more comfortable and confident when using the product.



## Principles of Good UX Design

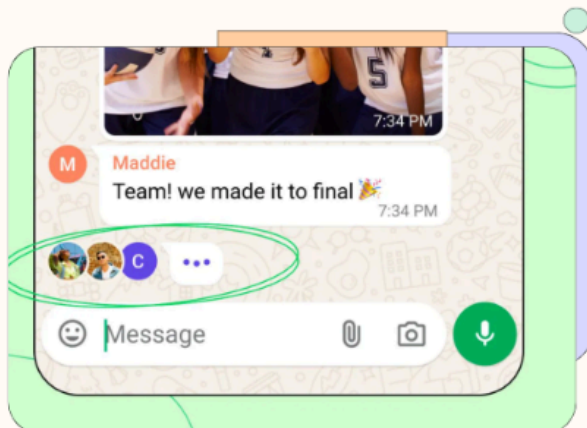
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**Users shouldn't have to relearn your product every time they open it.**

Apple's Human Interface Guidelines and Google's Material Design exist for exactly this reason — so that a “back” gesture, a share icon, or a settings gear behaves the same way across thousands of different apps. This is why you can pick up almost any new app and instinctively know where the menu or settings are — you're relying on cross-app consistency, not just consistency within one product.

4. **Feedback and Response:** Providing timely feedback is essential to keep users informed of their actions and the system's state. Whether it's a confirmation message, an error alert, or a loading indicator, feedback ensures that users know their inputs have been registered, and they understand the outcomes.



## Principles of Good UX Design

**Feedback and Response**

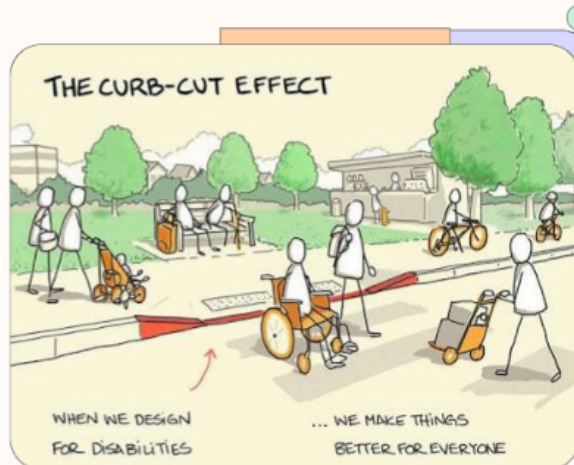
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**Every action needs a visible reaction — otherwise users assume something failed.**

WhatsApp's typing indicator (".") — tells you the other person is responding, so you don't wonder if your message got ignored.

5. **Accessibility:** A good UX design must be inclusive, ensuring that people of all abilities can use the product. Accessibility means designing products that can be navigated and understood by individuals with disabilities, whether it's providing text alternatives for images, ensuring keyboard navigability, or using sufficient color contrast.





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## Principles of Good UX Design

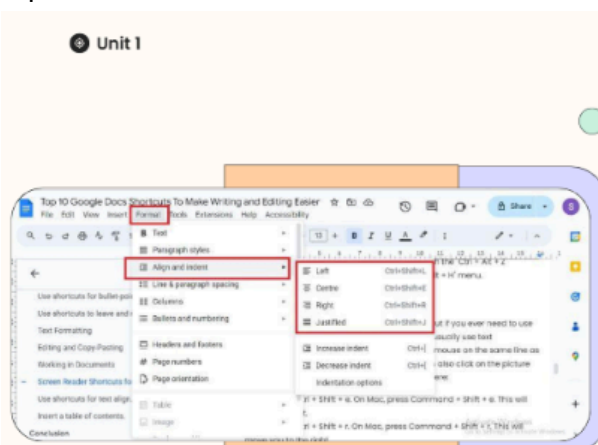
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**Good UX includes everyone, not just the "average" user.**

"Curb-Cut Effect": designing for the disabled often improves the experience for everyone. Same logic applies to captions (built for deaf users, but used by everyone watching a video on mute) or voice assistants (built for visually impaired users, now used by millions hands-free).

6. **Flexibility and Efficiency:** UX design should cater to both novice and experienced users. By providing shortcuts and customization options for advanced users while maintaining a simple interface for beginners, designers can create a flexible experience that meets the needs of all users.



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**Good design serves beginners and experts differently — without forcing either to suffer.**

Keyboard shortcuts. A beginner in Google Docs uses the menu to bold text. A power user just hits Ctrl+B. Same feature, two paths, both equally valid. Photoshop, Excel, and most professional software follow this exact principle — simple defaults visible on the surface, powerful shortcuts underneath for those who want speed.

7. **Error Prevention and Recovery:** Anticipating user errors and designing safeguards can prevent mistakes from occurring in the first place. When errors do happen, the design should offer clear instructions on how to recover, whether through undo options, helpful error messages, or guided assistance.



Unit 1
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## Error Prevention and Recovery

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**Assume users will make mistakes — design so mistakes are easy to fix.**

Gmail's "Undo Send" button. Instead of trying to make it impossible to send a bad email (which would be annoying and restrictive), Gmail gives you a 5–10 second window to undo it. It doesn't prevent the mistake — it makes the mistake reversible. Compare this to systems that show a scary "Are you sure?" popup for every single action — that's poor error design masquerading as caution; it trains users to blindly click "yes" anyway.

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## Key Elements of UX: Usability, Accessibility, Desirability, and Value

Successful UX design is made up of several key elements that work together to create a holistic user experience. These elements ensure that the product is not only functional but also enjoyable and meaningful to users.

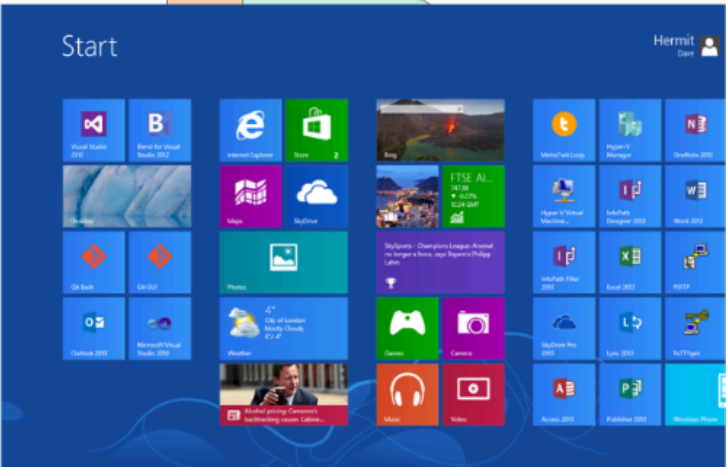
Let's explore the four key elements of UX: usability, accessibility, desirability, and value.

1. **Usability** is the backbone of good UX design. It refers to how easy and intuitive a product is to use. A usable product enables users to complete their tasks with minimal effort, confusion, or error. Usability focuses on:
  - **Learnability:** How easy is it for users to accomplish basic tasks when they first encounter the design?
  - **Efficiency:** How quickly can users perform tasks once they have learned the design?
  - **Memorability:** After a period of not using the product, how easily can users reestablish proficiency?
  - **Error Tolerance:** How well does the product prevent errors, and how easily can users recover from them?

- **Satisfaction:** How pleasant is it to use the product? A highly usable product meets these criteria, ensuring that users can interact with the system effortlessly and with satisfaction.

## • Usability Failure Windows 8

- Learnability** — Failed badly. Long-time Windows users couldn't find basic things like "Shut Down" (it was buried inside a charm menu you had to swipe in from the screen edge). Corporate IT departments had to run training sessions just to teach employees how to turn off their computers.
- Efficiency** — Worse for desktop users. Simple tasks that took one click in Windows 7 (open Start → find program) now required jumping between the tile screen and the traditional desktop.
- Memorability** — Poor. Users described constantly forgetting where features had moved, because nothing matched their existing mental model of Windows.
- Error tolerance** — Weak. There was no obvious "undo" for the confusion — no way back to the familiar Start menu at all in the initial release.
- Satisfaction** — Extremely low. Windows 8 became one of the most criticized OS releases in Microsoft's history, and enterprise adoption stalled hard, with many businesses skipping it entirely and staying on Windows 7.



Fix: Start button restored in 8.1, full Start menu back in Windows 10.

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- Accessibility** ensures that a product can be used by as many people as possible, regardless of their abilities or disabilities. It's not just about compliance with laws and regulations—it's about creating an inclusive experience for all users. Accessibility considerations include:
  - **Visual:** Ensuring that users with visual impairments can navigate and understand the content through features like screen reader compatibility, sufficient color contrast, and resizable text.
  - **Auditory:** Providing alternatives for sound-based information, such as captions for videos and alerts for individuals with hearing impairments.
  - **Motor:** Designing interfaces that are easy to navigate with different input methods, such as keyboards, adaptive devices, or voice controls.
  - **Cognitive:** Simplifying complex interactions and providing clear, understandable content for users with cognitive impairments. By prioritizing accessibility, UX designers ensure that their products are usable by everyone, making the experience more inclusive and widely adopted.
- Desirability** refers to the emotional response a product evokes in its users. It's about more than just functional performance—desirability influences how users feel about a product and whether they enjoy interacting with it. Elements that contribute to desirability include:

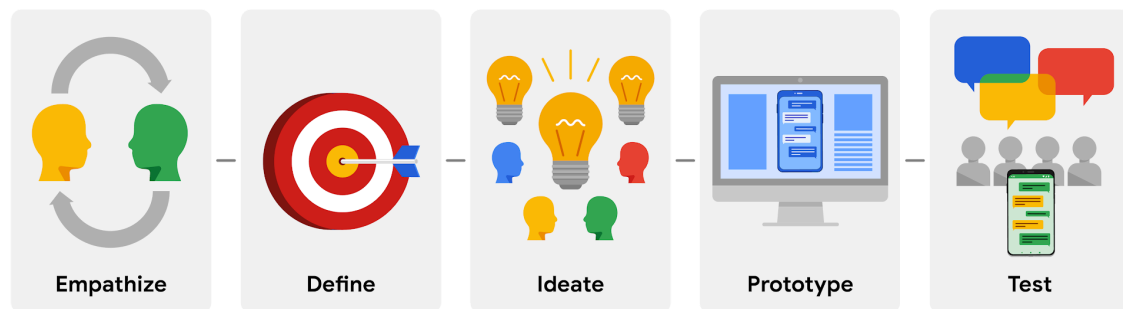
- **Aesthetic Appeal:** A visually attractive design can create a strong first impression and encourage continued use.
  - **Branding and Identity:** The way a product aligns with a user's personal values, aspirations, or identity can make it more desirable.
  - **Engagement and Fun:** Gamification elements, interactive features, and small moments of delight can add a sense of enjoyment to the experience. When users find a product desirable, they are more likely to use it frequently, recommend it to others, and develop a positive association with the brand.
4. **Value:** Ultimately, a product must deliver value to its users. This means that the product solves a real problem or fulfills a need in a way that is meaningful to the user. The value can be functional, emotional, or both, depending on the product and its purpose. Key questions that define value in UX design include:
- Does the product solve a real problem for the user?
  - Is the product worth the user's time, money, and attention?
  - Does the product improve the user's life in some way?
- A product that delivers value provides users with a reason to use it. This value, combined with strong usability, accessibility, and desirability, creates a compelling user experience that drives satisfaction and loyalty.

## Design Thinking: A UX Design Framework

In the world of UX design, a framework is a conceptual tool that provides guidance on the best practices and processes for solving problems and building solutions that solve the problems of real users. Frameworks provide structure for the design process and foster collaboration, which can spark innovations. Most UX designers follow a specific framework or process when approaching their work, from the first idea all the way through to the final launch of a product.

In this certificate program, you'll follow the design thinking framework to sequence the assignments that contribute to the designs that you'll create throughout the program. The design thinking framework is a user-centered approach to problem-solving that includes activities like research, prototyping, and testing to help you understand who your user is, what their problems are, and what your design should include.

The design thinking framework involves the following phases: empathize, define, ideate, prototype, and test.



Though it may sound like a linear process, the design thinking framework should be iterative, which means that you'll repeat certain phases as you refine your designs. For example, depending on the feedback you receive during testing, you might need to conduct additional research, brainstorm new ideas, or develop new prototypes.

Let's look at the five phases of the design thinking framework to learn more about which activities you'll perform during each one. Keep in mind that the design thinking framework as presented here is an idealized model for UX designers to follow, so you might see some variation in its implementation between different companies, teams, or projects.

## **Empathize**

During the empathize phase, your primary goal is to learn more about the user and their problems, wants, and needs, and the environment or context in which they'll experience your design. The most important part of the empathize phase is to step away from your assumptions and guesses and let your research findings inform your decision-making in later design phases.

Your user research might include user surveys, interviews, and observation sessions, and you might also need to conduct some research on the competitors' products to determine how your user frames competitors' products as part of their daily life and daily problem-solving.

## **Define**

In the define phase, you'll analyze your research findings from the empathize phase and determine which user problems are the most important ones to solve, and why. This will drive you toward a clear goal for the design of the product.

The most important outcome of this phase is a clear problem statement, which is a description of the user's need that your designs will address. You might also develop a value proposition, which is a summary of why your user would or should use the product or service that you're designing.

## **Ideate**

After you land on a user problem and establish why it's an important one to solve, it's time for the ideate phase. The goal of ideation is to come up with as many design solutions as possible—don't settle for your first solution because the most obvious solution is not always the right one.

Ideation involves collaborative brainstorming with other members of your team to generate as many solutions as possible to a problem. This could include marketing, engineering, product management, or any other stakeholders for the product or service. During brainstorming sessions, you should explore all possible solutions. Don't focus on whether something is a "good" or "bad" idea, just collect as many ideas as you can. The important thing here is to keep this process judgment-free.

After brainstorming, you'll then analyze your potential solutions and start to make choices about which ones are the best options to pursue as prototypes. You might return to user or competitive research to help you narrow down your ideas, and you might also create user flows to illustrate how the user will interact with your solution.

## **Prototype and Test**

After you have an idea of how to solve the problem, you're ready to enter the prototype phase, where your goal is to produce an early model of a product that demonstrates its functionality and can be used for testing. The test phase is critical to developing the right solution to address your user's problem, and an organized approach to testing can help you create exceptional user experiences.

Prototyping and testing are interconnected, which means that you'll test your designs at each stage of prototype development rather than waiting to test until after the working

prototype is complete. If the design is too polished the first time you present it to users, you might not get as much feedback. Think about ways to include testing throughout the design process, so that you're iterating your designs based on user feedback instead of other reasons.

For example, you might test the concepts behind your design by presenting users with a simple sketch, wireframe, or a sitemap. Taking what you learned, you might iterate on that design to a more detailed design on paper (known as a low-fidelity prototype) and conduct another round of user testing. At some point, you'll iterate the design again into a working, interactive model using a software program (also known as a high-fidelity prototype) and test that as well. You might also consider testing more than one prototype at the same time to get feedback on multiple solutions, or testing the same prototype on multiple platforms, such as a laptop, tablet, and smartphone.

The goal of testing prototypes is to continue to refine the prototype as you gain insight into whether the design for your product or service is easy to use and solves the user's problem. At some point, you'll finalize a prototype, and then you'll provide it to developers, who will then turn your design into a product.

## **Key takeaways**

The design thinking framework is only one type of framework that UX designers use to organize their approach to designs, often based on the product they're designing and the organization they're working for. No matter which frameworks you use in your career, they all have a few core principles in common:

- Focus on the user.
- Create solutions that address the user's problems.
- Collaborate with teammates across departments.
- Validate your designs.
- Iterate as needed to design the right user experience.

## **The UX Design Process**

The UX design process is a structured approach to creating products that provide meaningful and relevant experiences to users. It involves a series of stages, each with its own focus, that work together to ensure that the final product is user-centered, functional, and delightful to interact with. The UX design process is not linear; it is iterative, meaning

that insights from one stage can influence earlier stages, and adjustments are made as new information becomes available. At its core, the UX design process is about solving user problems. It begins with a deep understanding of the users' needs, continues through the creation of design solutions, and ends with the implementation of the final product. Each stage builds upon the last, with the aim of refining the design and ensuring that it meets user expectations. The key stages in the UX design process include: Research, Design, Prototyping, Testing, and Implementation.

## Stages of the UX Design Process:

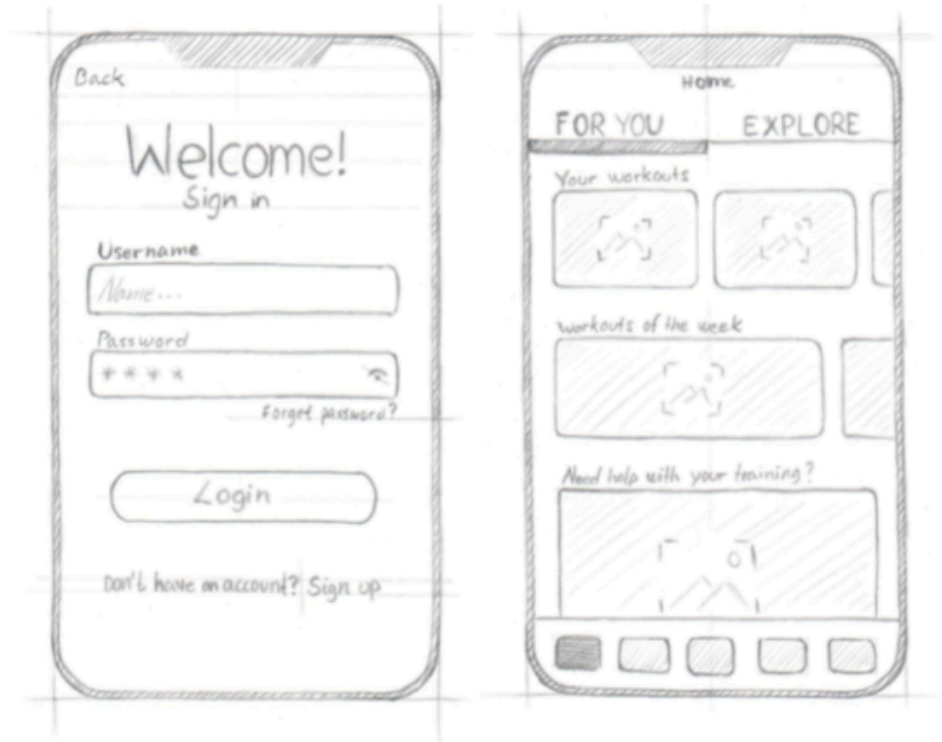
1. **Research:** The first stage in the UX design process is research, and it is arguably the most critical. This stage involves gathering insights about the target users, their needs, behaviors, motivations, and pain points. The goal is to develop a deep understanding of the people who will be using the product so that the design solutions can be tailored to meet their specific needs. There are several research methods UX designers use to collect data, including:
  - a. **User Interviews:** Speaking directly with users to understand their experiences, challenges, and preferences.
  - b. **Surveys and Questionnaires:** Gathering quantitative data from a large audience to identify trends and common issues.
  - c. **Contextual Inquiry:** Observing users in their natural environment to see how they interact with the product or service in real-time.
  - d. **Competitive Analysis:** Analyzing competitors' products to understand their strengths and weaknesses and identify opportunities for improvement.
  - e. **Analytics and Metrics:** Reviewing data from existing products to see how users currently engage with the system.

Research is foundational because it informs every subsequent decision in the design process. The insights gathered during this stage guide the design direction and ensure that the product is built with the user's needs in mind.

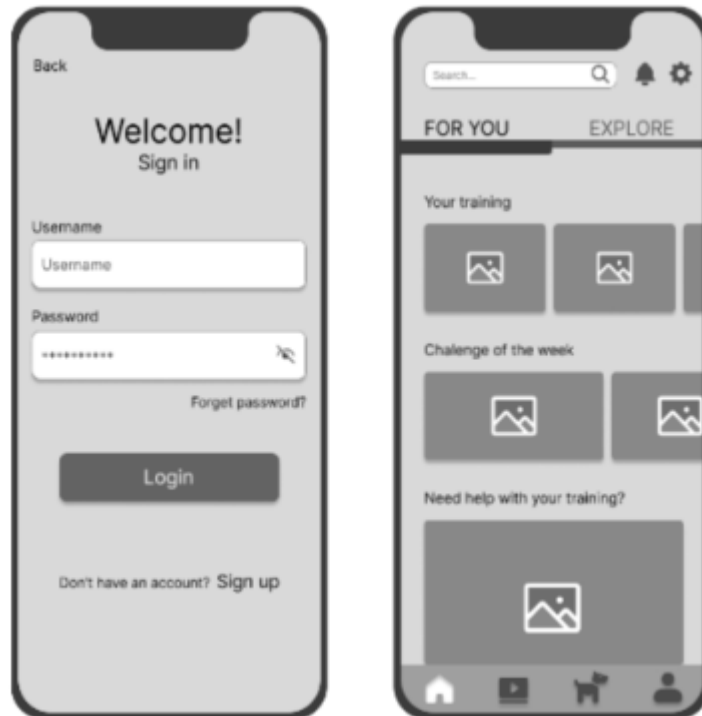
2. **Design:** Once the research phase is complete and the team has a strong understanding of the user, the design stage begins. This is where the creative solutions are generated, often starting with conceptual designs that address the user's needs and goals. The design stage is iterative, and it typically progresses through several phases:



- a. **Sketching and Ideation:** In the early stages of design, ideas are sketched out quickly and roughly. Designers brainstorm potential solutions, creating low-fidelity sketches that represent the basic structure of the product.



- b. **Wireframing:** Wireframes are simple, two-dimensional layouts that map out the product's structure without focusing on visual design. They represent the placement of key elements like navigation, content, and interactive components.



- c. **Information Architecture:** This involves organizing and structuring the content of the product in a way that makes it easy for users to find the information they need. It includes creating sitemaps, hierarchies, and taxonomies that guide the user's journey through the product.

### 1. Sitemap

This represents the main sections of the app and sub-sections within each.

#### Home

- Dashboard (Shows daily overview with progress ring, stats)
- Quick Stats (Steps, Calories, Active Minutes, Heart Rate)

#### Workout

- Browse Workouts (Categories: Cardio, Strength, Yoga, etc.)
- My Workouts (Saved or Recently Used)
- Active Workout (Displays workout in progress)

#### Progress

- Weekly Progress (Graph of activity over the week)
- Achievements & Badges (List of badges or achievements unlocked)
- Workout History (Log of past workouts)

#### Settings

- Profile Settings (Name, Age, Weight, etc.)
- Notifications (Enable/disable reminders, daily goals)
- Goal Settings (Set targets for steps, calories, active minutes)

## 2. Hierarchies

This hierarchy provides an outline of key screens and how they connect:

### Top Level

- Home
- Workout
- Progress
- Settings

### Second Level

#### - Home:

- Quick Stats, Daily Overview

#### - Workout:

- Browse Workouts → Category Listings → Individual Workout
- My Workouts
- Active Workout

#### - Progress:

- Weekly Progress Graph
- Achievements & Badges
- Workout History

#### - Settings:

- Profile Settings
- Notifications
- Goal Settings

## 3. Taxonomies

Let's break down content types and categorization.

### Workouts

- Categories: Cardio, Strength, Flexibility, Yoga, HIIT, etc.
- Attributes: Duration (15 min, 30 min, etc.), Difficulty Level (Beginner, Intermediate, Advanced)
- Status: Completed, In Progress, Saved

### Progress Metrics

- Time Periods: Daily, Weekly, Monthly
- Metrics: Steps, Calories, Distance, Heart Rate, Active Minutes

### Achievements & Badges

- Types: Milestones (e.g., 10,000 steps), Consistency (e.g., weekly streaks), Performance (e.g., longest workout)
- Levels: Bronze, Silver, Gold, Platinum (to encourage users to reach higher levels)

## User Journey Guide

### 1. Onboarding

- Users input basic profile information in Settings (age, weight, fitness goals).
- Users set up notification preferences and initial goals.

### 2. Daily Engagement

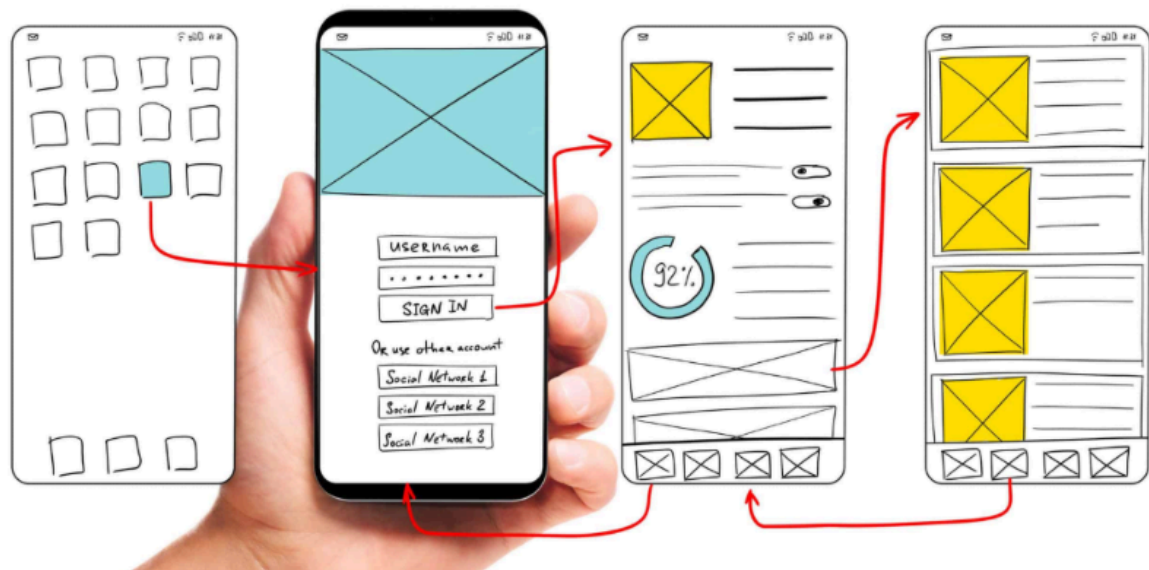
- Users access Home to view daily progress and navigate to Workout for a workout session.
- After the workout, users can check their **Progress** for achievements and review their history.

### 3. Long-term Engagement

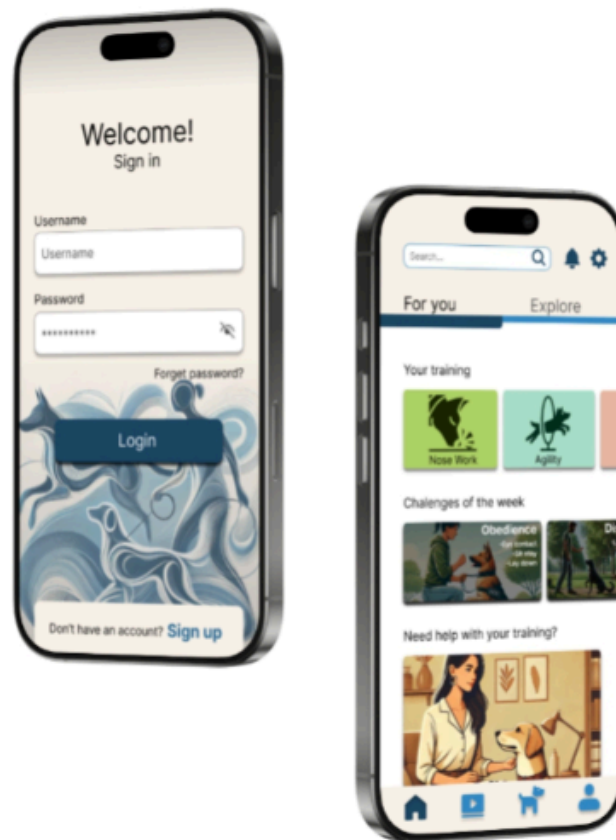
- Users explore Progress regularly to track progress.
- Settings allow them to update goals as they improve, helping maintain motivation.
- Achievements & Badges add an element of gamification, motivating users to stay consistent.

*A foundational Information Architecture (IA) for the fitness app, organizing content to make navigation intuitive and to guide the user seamlessly through their fitness journey.*

- d. **Interaction Design:** At this stage, designers focus on how users will interact with the product, defining the flow of tasks, user journeys, and the behavior of interactive elements like buttons, forms, and menus.



- e. **Visual Design:** Finally, the visual design phase brings the product to life with color, typography, imagery, and branding elements. This stage finalizes the product's aesthetics, creating an attractive and cohesive design.



Throughout the design stage, UX designers collaborate with other team members, such as product managers, developers, and stakeholders, to ensure that the design aligns with business goals and technical constraints while meeting user needs.

3. **Prototyping:** Once the design is in a good place, it's time to create prototypes. A prototype is an early sample or model of the product that simulates its functionality. Prototyping is essential because it allows designers to test their ideas before moving into full development.

Prototypes can vary in fidelity:

- a. **Low-Fidelity Prototypes:** These are simple, often paper-based models or basic clickable wireframes. They provide a rough idea of how the product will work and are useful for early-stage testing. It could be created using simple paper based prototypes or could use digital tools like Balsamiq, Figma, Sketch etc. to create basic wireframes.
- b. **High-Fidelity Prototypes:** These are more advanced, interactive models that closely resemble the final product in both design and functionality. High-fidelity prototypes allow for a more accurate representation of the user experience and are often used in later stages of testing.

Prototypes serve as a crucial tool for identifying potential issues early in the process. By building a prototype, designers can explore different design solutions, test interactions, and refine the user flow before investing time and resources in development.

Borcelle

9:41

# Prototype Fidelity – Low vs. High

Low-fidelity tests the idea. High-fidelity tests the experience

## Low Fidelity Prototype

- Paper models or basic clickable wireframes
- Rough idea of how it works
- Tools: paper, Balsamiq, basic Figma
- Best for early-stage testing

## High Fidelity Prototype

- Advanced, interactive models
- Closely resembles final product – look & function
- Tools: Figma, Sketch (advanced)
- Best for later-stage, accurate UX testing

9:41

3726, Ashton Lane, Austin

Hey John Doe,

What can we help you with today?

Plumber, HVAC...

New Offers

Get 50%OFF On Any First Service

Services

Plumbing Services

HVAC Services

Electric Services

9:41

SEARCH FLIGHTS

ROUND TRIP ONE WAY

From Select a city

To Select a city

Departure 02/04/20

Return 06/04/20

Passengers 1 Adult

Tourist class Business class

Search Flights

Search Booking guide Check-in More

9:41

SEARCH FLIGHTS

ROUND TRIP ONE WAY

From Select a city

To Select a city

Departure 03/03/2021

Return 03/10/2021

Passengers 1 Adult

Tourist class Business class

Search flights

Search Booking guide Check-in More

09:21

SEARCH FLIGHTS

ROUND TRIP ONE WAY

From Select a city

To Select a city

Departure 02/10/2021

Return 02/10/2021

Passengers 1 Adult

Tourist class Business class

Search flights

Search Booking guide Check-in More

4. **Testing:** Testing is a critical stage in the UX design process, as it allows designers to validate their designs with real users. The goal of testing is to identify usability issues, gather feedback, and make iterative improvements to the design.

There are several types of usability testing methods, including:

- a. **Usability Testing:** Users are asked to complete tasks using the prototype while being observed by the design team. This helps identify areas where users struggle, revealing issues with navigation, comprehension, or interaction.
- b. **A/B Testing:** Two or more variations of a design are presented to users to see which one performs better. This method is useful for testing specific elements, such as call-to-action buttons or layout changes.
- c. **Surveys and Feedback Forms:** Users may be asked to complete surveys or provide feedback about their experience after interacting with the prototype. This helps gather insights into user satisfaction and the product's perceived value.

Pro tip: You can sign up and use online tools like Microsoft Forms or Typeform.com to create surveys and polls.

Testing is an iterative process. Based on the feedback collected, designers return to the design stage to make improvements and then test again. This cycle continues until the product meets usability standards and satisfies user needs.

5. **Implementation:** The final stage in the UX design process is implementation. This is where the product is built and brought to life. The design is handed over to developers, who code and build the product according to the specifications provided by the UX and UI designers. During implementation, designers often work closely with developers to ensure that the design is executed as intended. Implementation also involves ensuring that the product is responsive, performs well across different devices, and meets accessibility standards. As the product is built, it's crucial to maintain open communication between the design and development teams to address any issues that arise during the development phase. In some cases, additional rounds of testing may be necessary during or after implementation, particularly if unexpected usability issues arise. Once the product is built and tested, it is finally ready for launch.

The UX design process is a dynamic and iterative journey that takes a product from initial research through to implementation. Each stage—**Research, Design, Prototyping,**

**Testing, and Implementation**—builds upon the last, ensuring that the final product not only meets the user's needs but also provides a satisfying, efficient, and enjoyable experience. By following a structured UX design process, designers can create products that are both functional and delightful, ultimately driving user satisfaction, engagement, and success.

## Design Sprints:

The Design Sprint is a proven methodology for solving problems through designing, prototyping, and testing ideas with users. Design Sprints quickly align teams under a shared vision with clearly defined goals and deliverables. Ultimately, it is a tool for developing a hypothesis, prototyping an idea, and testing it rapidly with as little investment as possible in as real an environment as possible.

A design sprint is:

- A time-bound process with five phases: Understand, Ideate, Decide, Prototype, and Test
- A specific implementation plan for solving a specific design challenge
- Strictly timebound from start to finish, typically spread over five full business days
- One method with a defined process

### Design Sprint Phases Overview

- The five phases are: understand, ideate, decide, prototype, and test, typically completed in five days.
- Understand: The team learns about the design challenge through expert insights and cross-department discussions, focusing on the user's needs.
- Ideate: Team members generate and sketch ideas, while also beginning to recruit users for testing.
- Decide: The team selects the most promising solution and plans the prototype.
- Prototype: A realistic, testable version of the solution is built, focusing on user experience.
- Test: The prototype is tested with users to gather feedback and identify necessary improvements.

### Outcome and Importance

- The sprint results in clear insights on what to adjust before launch.



- Design sprints are a valuable method in UX design for quickly iterating and validating ideas with users.

## Important Questions:

1. What is the difference between UI and UX?
2. What is a Design Sprint? Explain the stages involved with an example.
3. Describe the four key elements of UX (usability, accessibility, desirability, value) with one example each.
4. Explain the importance of accessibility in UX design.
5. Explain the complete UX Design Process in detail. Apply this process to outline how you would design a food delivery app from scratch.
6. Discuss the key principles of good UX design in detail. Choose any mobile app you use regularly and evaluate it against each principle, identifying strengths and areas for improvement.
7. Describe the Research stage of the UX Design Process in detail. Explain any four research methods used by UX designers to gather user insights.
8. Describe all five stages of the UX Design Process, explaining the key deliverables of each stage.
9. Differentiate between Low-Fidelity and High-Fidelity prototypes. Explain why prototyping is considered essential before moving into full development.
10. Explain the Testing stage of the UX Design Process. Describe the three usability testing methods and their purpose.

## CHAPTER II: UX Research

### Chapter II Agenda:

1. Introduction to User Research.

2. Types of User Research : Qualitative and Quantitative.
3. Integrating research into the Design Process
4. User centered design
5. Choosing the right research method
6. Bias in UX Research.
7. Translating research into actionable insights.

## Introduction to User Research:

User research is the cornerstone of any successful user-centered design process. It involves studying users' needs, behaviors, and pain points to inform design decisions and create functional and meaningful products. Understanding the user is crucial because it ensures that the product solves real problems and provides value in intuitive and enjoyable ways. Design teams risk creating products that miss the mark, frustrate users, or fail to meet their expectations without research. The role of user research in informing design decisions is indispensable. It drives decision-making by providing data and insights that help designers create solutions grounded in reality. Whether you're determining the layout of an app, the flow of a user journey, or the functionality of a new feature, user research ensures that these decisions are based on actual user needs rather than assumptions. By identifying pain points and understanding user behaviors, designers can prioritize features and refine designs to better serve their audience.

## Types of User Research:

User research can be broadly categorized into two main types:

- **Qualitative** Research and
- **Quantitative** Research.

Both types offer unique insights and are often used together to paint a complete picture of user behavior.

1. **Qualitative Research:** Qualitative research is exploratory and focuses on understanding the "why" behind user behaviors. It provides deep, nuanced insights into user motivations, preferences, and challenges.

**In-Depth Interviews:** In-depth interviews involve one-on-one conversations between the researcher and the user. These interviews aim to explore the user's thoughts, feelings, and experiences in great detail. They are particularly useful for uncovering insights that users may not be able to express through surveys or other quantitative methods.

**Techniques for Conducting Effective Interviews:** Please prepare open-ended questions that encourage the user to go over their experiences.

- Actively listen and probe deeper into interesting or unexpected responses.
- Maintain a neutral tone and avoid leading questions to ensure unbiased feedback.

Sample Questions:

- “Can you describe a recent experience using [product]?”
- “What challenges did you face while completing [task]?”
- “How did you feel about [specific feature], and why?”

**Ethnographic Studies:** Ethnographic studies involve observing users in their natural environments. This method allows researchers to gain insights into real-world usage and uncover behaviors that users might not articulate during interviews. Ethnography is especially valuable for understanding how products fit into users' daily lives and identifying contextual factors that influence their interactions.

Case Studies: Example: A healthcare designer observes nurses in a hospital setting to understand how they interact with electronic health records systems, identifying inefficiencies and pain points in real-time workflows.

**Focus Groups:** These groups involve guided discussions with small groups of users. The researcher facilitates the sessions, asking questions and encouraging participant interaction. The groups are influential in gathering diverse perspectives, generating ideas, and exploring user reactions to specific features or concepts.

**Facilitating Group Discussions:** Start with broad questions to foster open discussion, then narrow the focus to specific topics. Balance the conversation by encouraging quieter participants to share their thoughts. Use visual aids or prototypes to stimulate conversation and get feedback on specific design elements.



## Methods

- In-Depth Interviews**
  - ONE-ON-ONE CONVERSATIONS BETWEEN RESEARCHER AND USER
  - Explore thoughts, feelings, and experiences in depth
  - Uncover insights users can't easily express in a survey
  - Best for the "why" behind behavior — not the "how many"
- Sample Questions:**
  - "Can you describe a recent experience using [product]?"
  - "What challenges did you face while completing [task]?"
  - "How did you feel about [specific feature], and why?"
- Ethnographic Studies**
  - Observing users in their natural environment
  - Reveals real-world usage, not reported usage
  - Surfaces behaviors users can't put into words
  - Shows how a product fits into daily life & context
- Focus Groups**
  - Guided discussion with a small group of users
  - Researcher facilitates, encourages interaction
  - Great for diverse perspectives & idea generation
  - Useful for reactions to specific features or concepts
- Facilitating Group Discussions**
  - Start broad, then narrow to specific topics
  - Draw out quieter participants — balance the conversation
  - Use prototypes or visuals to spark concrete feedback

- Quantitative Research:** Quantitative research involves collecting numerical data that can be analyzed statistically. It helps identify patterns, measure user behavior at scale, and test hypotheses.

**Surveys:** Surveys are a common method for collecting quantitative data from a large audience. They are useful for gathering broad insights into user preferences, attitudes, and behaviors.

**Designing Effective Surveys:** Keep the survey short and focused to reduce drop-off rates. Use a mix of closed-ended and open-ended questions to balance quantitative data with qualitative insights. Avoid leading questions and ensure that answer options are mutually exclusive and exhaustive.

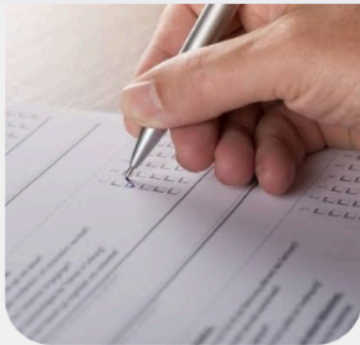
#### Common Pitfalls:

- Survey fatigue:** Asking too many questions can overwhelm participants.
- Ambiguity:** Poorly worded or vague questions can lead to inaccurate data.
- Bias:** Leading questions or unbalanced answer options can skew results.

**Analytics:** Analytics involves tracking user interactions on websites or apps to gather data on how users behave within the product. Metrics such as page views, click-through rates, and conversion rates provide insights into user engagement and help identify areas for improvement.

**Using Web and App Analytics:** Set up tracking for key performance indicators (KPIs) such as bounce rates, session duration, and conversion goals. Analyze user flows to see where users drop off and which paths lead to successful outcomes. Use heatmaps and click tracking to identify areas of high engagement or friction.

**Interpreting Metrics:** Look for patterns in the data that align with user pain points or behaviors identified in qualitative research. Use A/B testing to experiment with different design variations and measure their impact on user behavior.



#### ○ Surveys

Surveys are a common method for collecting quantitative data from a large audience. They are useful for gathering broad insights into user preferences, attitudes, and behaviors.

#### ○ Designing Effective Surveys

- Keep the survey short and focused to reduce drop-off rates.
- Use a mix of closed-ended and open-ended questions to balance quantitative data with qualitative insights.
- Avoid leading questions and ensure that answer options are mutually exclusive and exhaustive.

#### ○ Analytics:

- Tracks real user interactions on a website or app
- Key metrics: **page views**, **click-through rate**, **conversion rate**
- Reveals engagement patterns and problem areas at scale

#### ○ Interpreting Metrics

- Cross-check patterns against qualitative findings
- Run A/B tests to measure the impact of design changes

#### ○ Using Web & App Analytics

- Track KPIs — bounce rate, session duration, conversion goals
- Map user flows to spot drop-off points
- Use heatmaps & click tracking to find friction or high engagement

|                   | Qualitative                           | Quantitative                  |
|-------------------|---------------------------------------|-------------------------------|
| Core question     | Why does it happen                    | How many, how often           |
| Sample size       | Small, 5–15 people                    | Large, 100+ people            |
| Data type         | Words, quotes, video                  | Numbers, statistics           |
| Methods           | Interviews, ethnography, focus groups | Surveys, analytics, A/B tests |
| Analysis          | Finding themes and patterns           | Statistical analysis          |
| Flexibility       | Can adapt mid-conversation            | Fixed, set before launch      |
| Time & cost       | Slower, resource-heavy per person     | Faster and cheaper to scale   |
| Researcher's role | Active listener and interpreter       | Neutral collector of data     |
| Output            | Narratives, quotes, themes            | Charts, percentages, trends   |
| Best for          | Understanding motivation & context    | Measuring behaviour at scale  |

## Integrating research into the Design Process

In modern product development, design without research relies heavily on designer bias, unsubstantiated assumptions, and intuition. **Research-driven design** shifts product development from an aesthetic exercise to an *evidence-based problem-solving discipline*.

The core purpose of research integration is to ground every design decision—from high-level architecture down to micro-interactions—in verified user behaviors, mental models, and environmental constraints.

**Key Concept: Evidence-Based Design** An approach where design decisions are explicitly justified by empirical qualitative and quantitative user research rather than personal preference or organizational hierarchy.

### Case Study Framework: The "BeanExpress" Application

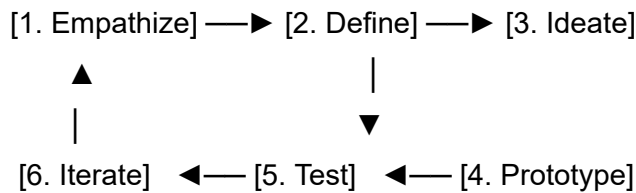
To examine how research operates across the design lifecycle, consider the primary case study below.

#### Case Study: BeanExpress

**Context:** *BeanExpress* is a mobile ordering application for a high-volume local coffee chain. **Business Problem:** The app suffers from a **42% order**

**abandonment rate** during peak morning hours (7:00 AM – 9:00 AM). **Goal:** Increase order completion rates by aligning the app experience with commuter behavior.

### The Six Stages of Research Integration



#### Stage 1: Empathize

The *Empathize* stage focuses on unearthing true user behaviors, environmental context, and emotional friction points through direct observation and engagement.

- **Primary Methods:** Contextual inquiry, semi-structured user interviews, direct observation (shadowing).
- **Case Study Application:** Researchers observe 12 commuters using the current *BeanExpress* app while walking to urban train stations.
- **Key Finding:** 8 out of 12 users struggled to customize milk choices because the option was buried inside a multi-level nested dropdown menu. Users canceled their order out of frustration or missed their train while attempting to navigate the interface.

#### Stage 2: Define

In the *Define* stage, raw research data is synthesized into structured artifacts to establish clear boundaries for the problem space.

- **Primary Artifacts:** User Personas, Empathy Maps, Problem Statements, and *How Might We* (HMW) frameworks.
- **Case Study Application:**
  - **User Persona:** *Commuter Clara* — A time-sensitive professional needing her specific coffee order placed in under 15 seconds while walking through crowded environments.
  - **Problem Statement:** "Time-constrained morning commuters need a streamlined method to re-order customized beverages in three taps or fewer because multi-step menu navigation leads to high cognitive load and order abandonment."

### Stage 3: Ideate

The *Ideate* stage leverages research-backed problem statements to generate a wide range of targeted structural and visual solutions.

- **Primary Techniques:** "Crazy Eights" sketching, feature prioritization matrices, rapid concept mapping.
- **Case Study Application:** Rather than adding low-priority features (e.g., social sharing or gamified loyalty badges), the design team focuses exclusively on low-friction checkout flows. Ideas include a homescreen "One-Tap Usual" widget and quick-toggle customization drawers.

### Stage 4: Prototype

The *Prototype* stage converts conceptual ideas into tangible artifacts that can be tested with real humans prior to software engineering.

- **Primary Output:** Low-fidelity paper wireframes, interactive screen flows (e.g., Figma wireframes).
- **Case Study Application:** Designers construct a interactive low-fidelity wireframe featuring a prominent "**Re-Order My Usual**" card directly on the home screen, complete with instant toggles for milk type and temperature.

### Stage 5: Test

The *Test* stage systematically evaluates prototype performance against user expectations and benchmark metrics.

- **Primary Metrics:** Task completion rate, time-on-task, error rate, System Usability Scale (SUS) scores.
- **Case Study Application:** Usability tests are conducted with 8 target users assigned the task: *"Order your custom oat-milk latte using the prototype while walking."*
- **Observations:** Average order completion time drops from **45 seconds** to **9 seconds**. However, 3 participants misidentified the "Quick Re-Order" card as a promotional advertisement due to its full-width banner layout.

### Stage 6: Iterate

The *Iterate* stage uses empirical feedback from the *Test* stage to refine designs continuously.



- **Primary Activity:** Usability severity scoring, design revision, secondary validation testing.
- **Case Study Application:** Based on testing feedback, the team redesigns the "Quick Re-Order" module from a banner layout into a floating action card with clear transactional typography. Re-testing verifies 100% recognition and completion success.

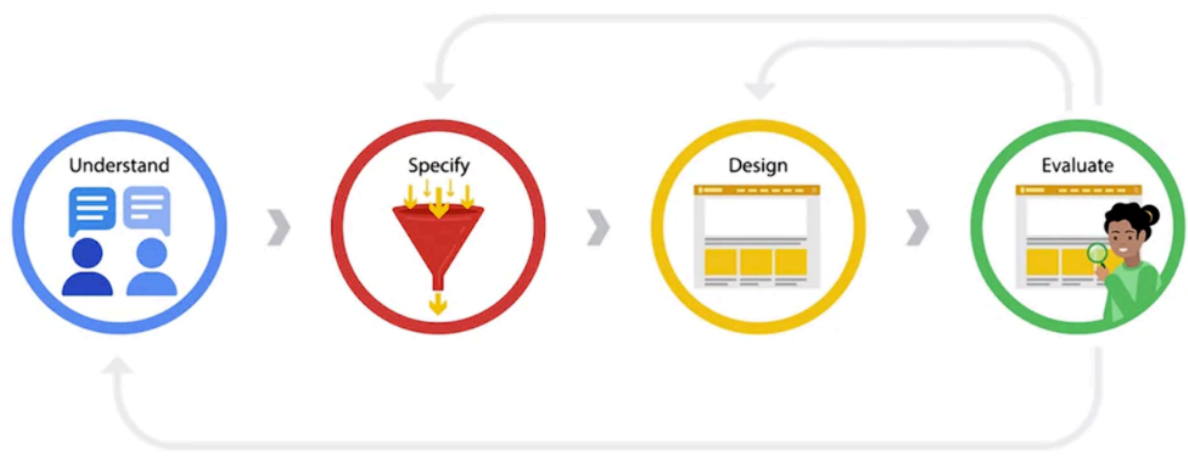
### Strategic Benefits of Integrating User Research into the design process:

| Benefit                                | Mechanism                                                                       | Practical Outcome in Case Study                                |
|----------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Bias Reduction</b>                  | Replaces internal opinions with observed user data.                             | Avoided spending \$50,000 developing unwanted social features. |
| <b>Usability Enhancement</b>           | Directly removes identified friction points in critical user flows.             | Reduced average order completion time by <b>80%</b> .          |
| <b>Risk Mitigation</b>                 | Identifies usability flaws in low-fidelity prototypes prior to code production. | Identified layout confusion before backend API integration.    |
| <b>Inclusivity &amp; Accessibility</b> | Integrates varied user constraints (e.g., one-handed use while walking).        | Ensured high-contrast UI and single-thumb touch targets.       |

## Good to Know : User Centered Design

Why is it important that designs are centered on the user? Because the user buys and uses your product. In addition, it's also important to solve problems that people actually experience, rather than only trying to solve problems you personally experience. Focusing on real user problems reduces the impact of designer bias. Larry Page, one of Google's

founders, highlighted just how important user-centered design is when he said, "There is no substitute for personally watching and listening to real people." Focus on the user and all else will follow. User-centered design puts the user front and center. Focusing on the user means considering their story, emotions, and the insights you've gathered about them. To keep our focus on the user, the user-centered design process has four steps: understand, specify, design, and evaluate. Following this framework helps us build products people actually want to use. First, understand how the user experiences the product or similar products.



Really understanding the end user requires a lot of research. Next, specify the end user's needs. Based on your research, you'll narrow down which end-user problem is the most important to solve. Then, design solutions to the end user's problem. This is where you'll come up with ideas for what the product might look like and actually start building the product. Finally, evaluate your design against your end user's needs. Does your design solve the end user's problem? You'll find out here, and you'll also do it by testing your product with real people. It's important to keep in mind that as you go through this process, iteration is key. Iteration means doing something again, by building on previous versions and making tweaks. Let's think about an example of user-centered design in the real world. Five years ago, Google launched Google Photos as a place to store photos and keep them organized. Over time, they noticed that users also wanted to revisit memories and relive treasured moments. To make it easier for users to do this, they use the user-centered design process to redesign Google Photos. Now users can easily search to find their memories. Now you understand the importance of user-centered design and why designers always keep the user front and center.

In the field of user experience design, the "user" comes first. A user is a person who is trying to solve a problem and is looking for a product or service to help them solve it. The user

experience is the journey that the user takes with that product or service. As a UX designer, your goal is to keep the user at the center of every decision you make, and to do that, you need to get to know your user.

Most of the initial UX research that you conduct at the beginning of the project will be focused on getting to know the characteristics of your users, their goals, and their pain points. Understanding the user empowers you to design experiences that are helpful or easy to use.

One of the key challenges of getting to know your users is avoiding taking a narrow view of the user, or making assumptions about what users need based on stereotypes. Users who seem to be different from one another might actually have shared wants and needs. Make sure to step back and get a clear picture of all your potential users—which often means that you'll need to specifically plan to address the needs of users who are often overlooked.

To reach as many users as possible, you'll consider questions such as these as part of your user research approach:

- Do my users have impairments or disabilities to consider—whether temporary, situational, or permanent?
- How familiar are my users with technology?
- How are my users accessing the product or service?
- Where and when are my users accessing the product or service?
- Have I considered all my potential users?

The goal for user research is to get the widest possible selection of potential users to include as research participants. Make sure to consider income level; demographic data such as age, gender, and ethnicity; educational background; and geographic location.

### **Do my users have impairments or disabilities to consider?**

You might be surprised to find out that more than 1 billion people around the world have a disability, out of a global population of 7.8 billion. In the U.S., one in four adults identifies as having a disability. The term accessibility refers to the design of products, devices, services, or environments for people with disabilities. As a UX designer, it's important to keep users with disabilities in mind as you design features or new products. Designing for accessibility isn't an obstacle, but a way to get your products to as many users as possible.

Accessibility is about making products accessible to all people, whether they have a disability or not. Think about inclusive product features that increase magnification, like enlarged fonts. Or features that help everyone understand videos, like closed captions.

### **How familiar are my users with technology?**

User experience doesn't just focus on the experience of existing users, however. It also includes paying attention to the needs of people who are about to become users. Nearly a billion people, of all ages, all around the world, are getting online for the very first time.

When it comes to designing for this emerging population of potential users, their level of digital literacy is a key consideration. The American Library Association defines digital literacy as "the ability to use information and communication technologies to find, evaluate, create, and communicate information, requiring both cognitive and technical skills." People with limited digital literacy might not be familiar with certain design patterns, calls to action, or icons that those of us with more online experience take for granted. For example, they might not know what "swipe" means in relation to a touchscreen. They may not even know what a touchscreen is. This can affect their confidence and willingness to explore new technology.

### **How are my users accessing the product or service?**

It's important to keep in mind that not all users will access your product or service in the same way due to the cost and availability of devices and internet services. These factors are often due to other broader and historical factors in the markets you might be designing for.

Some users might not have reliable or unlimited access to the internet. This might be because the data they have purchased has run out or because their network coverage is inconsistent or sporadic. As a UX designer, you should try to find ways to take the offline experience into account as well. For more examples of ways you can create a rich offline experience for your users, check out the article [Offline UX design guidelines](#) on [web.dev](#).

### **Where are my users accessing the product or service?**

In some cases, you might be designing a product or service intended for use by people in various locations around the world, which creates some additional considerations, like local languages and cultural norms. Keep in mind that cultures and norms are ever-changing as people integrate and recontextualize technology in their lives, so you'll need to continue to

research potential users to identify shifts in user problems, preferences, and usage patterns over time.

Users who speak languages other than the primary language of the product or service you're designing might want or need to switch languages on their device depending on what they're trying to accomplish. For example, a user might want to read in Hindi but type using the English keyboard. Or the user might encounter a concept that cannot be easily translated from one language to another. Designing a multilingual keyboard option, and using universally-recognized icons, like an icon with a local currency sign for a banking app, are just a few key ways UX designers can make it easier for users in different locales.

### **Key takeaway**

Your goal as a UX designer is to solve a user problem by creating a design that makes every user feel like you designed the experience just for them no matter who they are, where they live, how much they earn, what their abilities are, or how highly they're educated. Gaining a comprehensive understanding of your user's context is an important place to start, and you're already on your way.

## Choosing the right user research method: The UX Research Method Landscape

Choosing the right user research method is not about personal preference; it depends on **what kind of question you are trying to answer** and **what stage of development your product is in**.

To categorize research methods, UX researchers use a 3-dimensional framework originally formalized by Christian Rohrer (Nielsen Norman Group):



## The Three Fundamental Dimensions

### Dimension 1: Attitudinal vs. Behavioral

- **Attitudinal (What People Say):** Measures users' stated beliefs, perceptions, expectations, and self-reported feelings.
  - *Methods:* Self-reported surveys, interviews, focus groups.
  - *Primary Use:* Understanding mental models, brand sentiment, and declared needs.
  - *Caution:* What users say they do often drastically differs from what they *actually* do (due to biases like Social Desirability or Memory Decay).
- **Behavioral (What People Do):** Measures direct, observable actions taken by users while interacting with a product or prototype.
  - *Methods:* Unmoderated usability testing, eye-tracking, analytics tracking, A/B testing.
  - *Primary Use:* Validating actual task completion, finding workflow bottlenecks, and tracking conversion funnels.

### Dimension 2: Qualitative vs. Quantitative

- **Qualitative (Why & How to Fix):** Generates rich, non-numerical data through direct observation, open-ended questioning, and analyzing behaviors.
  - *Sample Size:* Small ( $N = 5$  to  $12$  per user segment).
  - *Core Question:* "Why are users getting confused, and how can we fix the design?"
- **Quantitative (How Many & How Much):** Generates numerical data that can be mathematically analyzed, aggregated, and benchmarked.
  - *Sample Size:* Large ( $N = 100+$  for statistical significance).
  - *Core Question:* "How many users fail this step, and how much better is Design A than Design B?"

### Dimension 3: Context of Use

- **Natural / Near-Natural Use:** Observing users in their actual environment using the product spontaneously (e.g., field studies, analytics).
- **Scripted / Task-Based:** Asking participants to execute specific scenarios in a controlled environment (e.g., benchmark usability testing).
- **Decontextualized:** Studying user mental structures without direct product interaction (e.g., card sorting for information architecture).

### Decision Matrix: Matching Goals to Methods

| Research Goal / Question                            | Best Method(s)                        | Primary Dimension         |
|-----------------------------------------------------|---------------------------------------|---------------------------|
| Understand user routines and real-world environment | Contextual Inquiry, Field Observation | Qualitative + Behavioral  |
| Discover user needs, pain points, and motivations   | Semi-structured Interviews            | Qualitative + Attitudinal |

|                                                      |                                           |                            |
|------------------------------------------------------|-------------------------------------------|----------------------------|
| <b>Evaluate navigation structures and taxonomy</b>   | Card Sorting, Tree Testing                | Qual/Quant + Attitudinal   |
| <b>Evaluate UI friction and task completion flow</b> | Usability Testing (Moderated/Unmoderated) | Qualitative + Behavioral   |
| <b>Measure satisfaction or NPS at scale</b>          | CSAT / SUS Surveys                        | Quantitative + Attitudinal |
| <b>Compare two layout variations for conversion</b>  | A/B Testing, Multivariate Testing         | Quantitative + Behavioral  |

### Best Practice: The Mixed-Methods Approach (Triangulation)

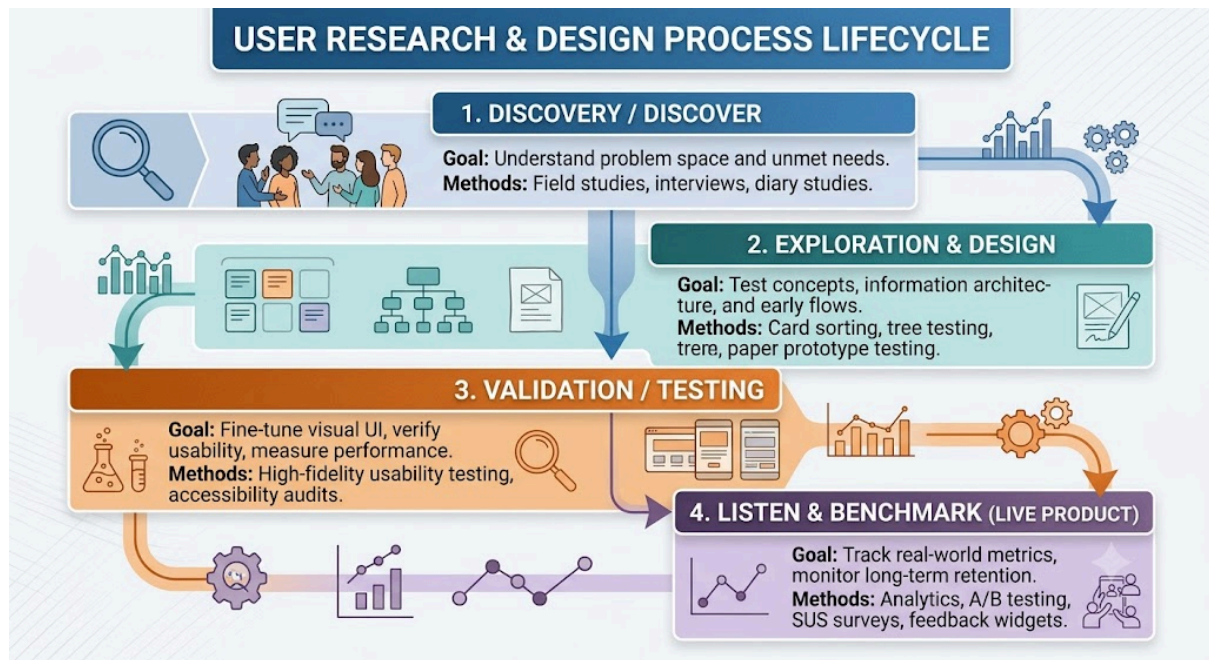
Relying on a single research method introduces systematic blind spots. The most robust user research utilizes **Triangulation**—combining qualitative and quantitative data sources to validate findings.

#### Example Triangulation Strategy:

1. **Analytics (Quantitative/Behavioral):** Reveals *where* drop-offs occur (40% drop-off at Step 2).
2. **Usability Testing (Qualitative/Behavioral):** Uncovers *why* drop-offs occur (users do not see the continuation button).
3. **Exit Survey (Quantitative/Attitudinal):** Measures *how widespread* user frustration is across the entire active user base.

### Choosing Methods Across Product Development Stages





Selecting research methods depends heavily on where a product sits in its development lifecycle. The four stages form a continuous feedback loop:

### 1. Discovery / Discover

- **Focus:** Uncovering the problem space before designing solutions.
- **Primary Goal:** Identify unmet user needs, contextual constraints, and pain points in existing workflows.
- **Key Methods:** Field studies, contextual inquiry, semi-structured interviews, and diary studies.

### 2. Exploration & Design

- **Focus:** Evaluating early concepts, mental models, and structural navigation.
- **Primary Goal:** Test core information architecture (IA) and user flows to ensure intuitive system layout.
- **Key Methods:** Card sorting, tree testing, wireframe walkthroughs, and paper prototype testing.

### 3. Validation / Testing

- **Focus:** Evaluating screen-level usability and interaction design before writing code.
- **Primary Goal:** Identify UI friction, verify task completion speed, and ensure accessibility compliance.

- **Key Methods:** High-fidelity interactive usability testing, task success tracking, and accessibility audits.

#### 4. Listen & Benchmark (Live Product)

- **Focus:** Monitoring real-world behavior and satisfaction post-launch.
- **Primary Goal:** Track quantitative usage metrics, validate feature adoption at scale, and gather continuous feedback.
- **Key Methods:** Web/app analytics, A/B testing, System Usability Scale (SUS) surveys, and in-app feedback widgets.

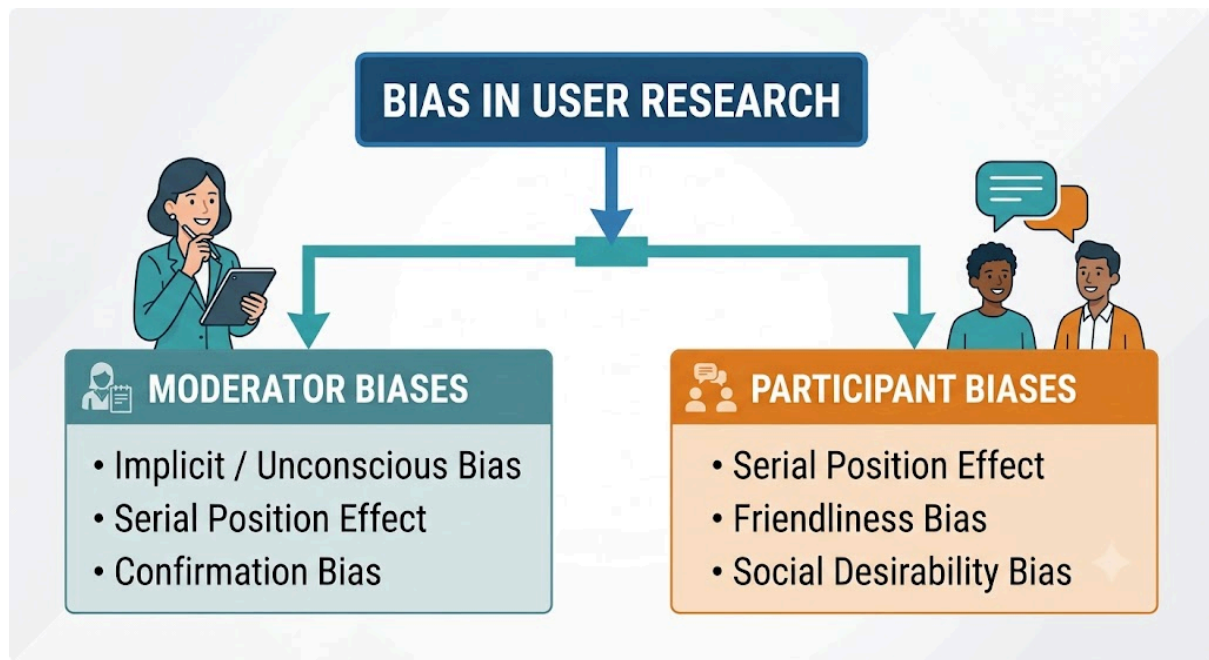
### Bias in UX Research:

All human beings are inherently susceptible to **cognitive and behavioral biases**—systematic patterns of deviation from norm or rationality in judgment. In user research, bias distorts data, compromises study validity, and leads to product decisions based on false assumptions rather than authentic user behavior.

Because many biases operate at an **unconscious level**, researchers often experience or trigger them without conscious awareness. The primary objective of rigorous research methodology is not to eliminate human bias entirely—which is impossible—but to **identify, recognize, and actively guard against it**.

### Taxonomy of Biases in User Research

Biases in user testing generally fall into two primary operational categories: **Moderator-Side Biases** (introduced by the researcher) and **Participant-Side Biases** (introduced by the subject).



## Moderator-Side Biases

### 1. Implicit (Unconscious) Bias

- **Definition:** The automatic association of attitudes, stereotypes, or expectations with specific groups of people without conscious awareness. These associations are frequently negative, exclusionary, or disempowering.
- **Impact on Research:** During participant recruitment and study execution, implicit bias can cause researchers to lower or alter expectations for certain user demographics, skewing how participant interactions are interpreted or evaluated.
- **Real-World Scenario:** A moderator unconsciously assumes an older participant will struggle with a smartphone app, leading the moderator to offer unsolicited hints or interpret minor hesitations as failure.

### 2. Confirmation Bias

- **Definition:** The tendency for researchers to search for, interpret, and remember information in a way that confirms their pre-existing hypotheses or design preferences.
- **Impact on Research:** The researcher highlights feedback that supports their preferred design variations while ignoring or rationalizing away negative feedback.

## Category B: Participant-Side Biases

### 1. Friendliness Bias (Acquiescence Bias)

- **Definition:** The tendency for participants to agree with or validate the researcher to avoid confrontation, maintain harmony, and remain polite.
- **Impact on Research:** Participants obscure genuine usability issues, withhold critical feedback, or claim a flawed feature is "easy to use" because they do not want to offend the moderator.
- **Real-World Analogy:** Agreeing to go to a friend's favorite restaurant and saying the food is great, even if you dislike it, simply to keep the interaction pleasant and non-confrontational.

## 2. Social Desirability Bias

- **Definition:** The tendency of participants to answer questions or perform tasks in a manner that will be viewed favorably by the researcher or peers, rather than reflecting their true behaviors or feelings.
- **Impact on Research:** Participants overstate positive habits (e.g., frequency of exercise, reading privacy policies) and underreport negative or embarrassing behaviors (e.g., skipping security checks, impulse buying).
- **Real-World Analogy:** Claiming to love a critically acclaimed movie when surrounded by a group of enthusiastic friends, even if you personally found it boring.

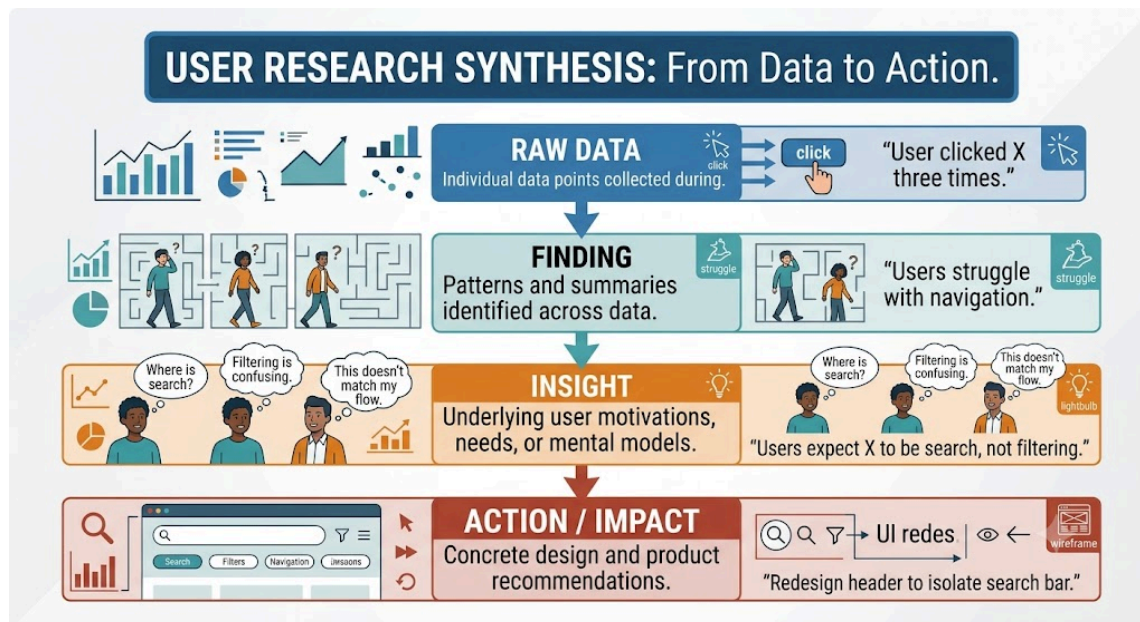
## Category C: Dual-Impact Biases (Affecting Both Parties)

### 1. Serial Position Effect

- **Definition:** A psychological phenomenon stating that when presented with a list or sequence of items/events, individuals best recall the **first items** (*Primacy Effect*) and the **last items** (*Recency Effect*), while middle items tend to blur.
- **Impact on the Moderator:** During analysis or long interviews, a researcher is far more likely to remember a participant's opening and closing remarks while forgetting critical feedback delivered mid-session.
- **Impact on the Participant:** When given multi-step instructions or long lists of options, participants primarily retain and act upon the first and last items communicated.

## Translating User Research into Actionable Insights

Raw data collected during user research is useless until it is processed and translated. To make research impactful, teams must elevate observations from mere facts to **actionable insights**.



**Finding:** A **Finding** is an objective, empirical summary of observed behaviors, recurring user actions, usability errors, or statistical metrics gathered during research. It answers the question: *"What actually happened during the study?"*

- **Descriptive and Objective:** It states facts without making assumptions about *why* the user acted that way.
- **Measurable or Pattern-Based:** Often backed by quantitative task completion rates, drop-off analytics, or qualitative counts (e.g., *"8 out of 10 participants..."*).
- **Surface-Level:** It describes the visible symptom rather than the root cause.

**Example:** *"60% of participants missed the 'Submit' button on the order confirmation screen."*

- **Do not treat findings as Insights:** Stopping at the finding stage often leads teams to apply "band-aid" design fixes (e.g., simply making the button bigger or making it bright red) without understanding why users missed it in the first place.

**Insight:** An **Insight** is the underlying psychological truth, mental model, frustration, or behavioral driver behind a finding. It answers the question: *"Why did the user behave this way, and what mental model caused it?"*

- **Explanatory and Root-Cause Focused:** Uncovers the cognitive friction, misaligned expectations, or contextual constraints driving the behavior.
- **Empathy-Driven:** Explains how the user perceives the interface or system based on past habits or mental shortcuts (heuristics).
- **Non-Prescriptive:** Defines the core problem space without dictating a rigid solution yet.

**Example:** *"Users perceive full-width banners at the bottom of a form as non-clickable visual advertisements (banner blindness), leading them to ignore the 'Submit' button placed inside that layout."*

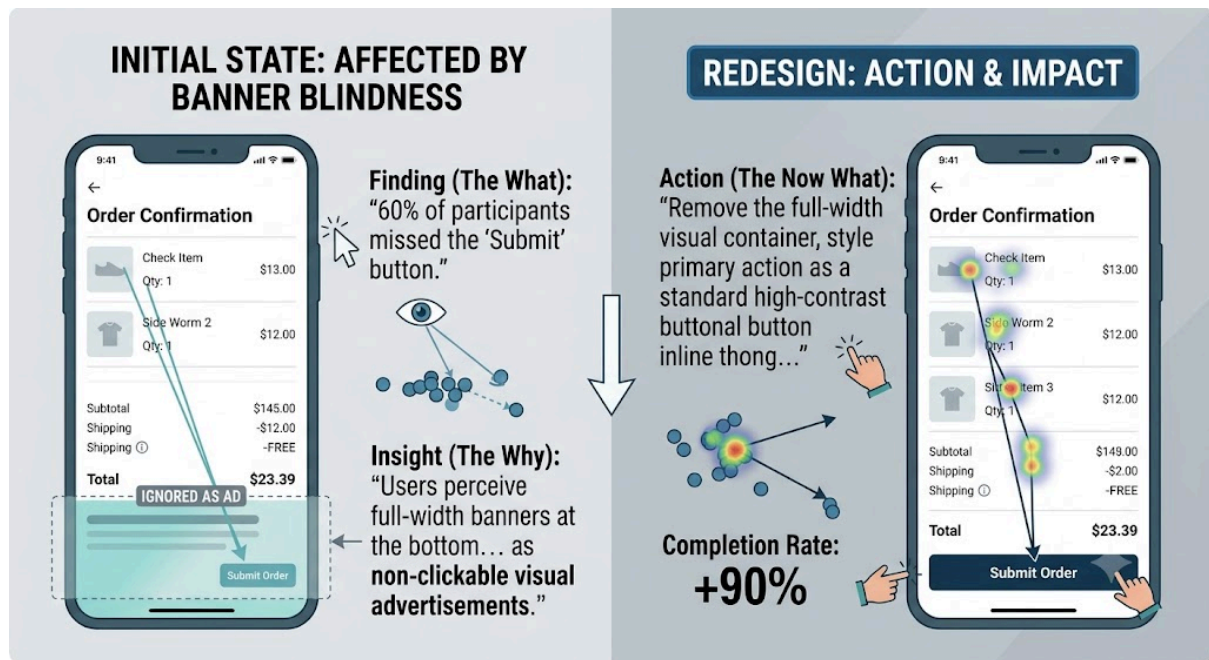
### How to Uncover Insights from Findings?

- **The "5 Whys" Technique:** Repeatedly question the observed behavior until you arrive at the fundamental user expectation.
- **Cognitive Walkthroughs:** Analyze the visual hierarchy and design patterns through established human psychology rules (e.g., Gestalt principles, banner blindness, cognitive load).

**Action:** An **Action** is the concrete, prioritized product recommendation, design modification, or strategic decision directly informed by the insight. It answers the question: *"What will we build, change, or remove to solve the underlying problem?"*

- **Solution-Oriented:** Translates psychological understanding into tangible engineering or UX tasks.
- **Testable and Measurable:** Defines clear acceptance criteria or success metrics (e.g., *"Redesign will reduce drop-off by X%"*).
- **Targeted at the Root Cause:** Fixes the foundational mental model clash identified in the insight stage, rather than patching symptoms.

**Example:** *"Remove the full-width visual container, style the primary action as a standard high-contrast button inline with the form fields, and sticky-pin it above the fold."*



## Important Questions:

1. Differentiate between qualitative and quantitative research methods. Discuss the core question each type answers, along with two example methods for each.
2. Explain the taxonomy of biases in UX research. Describe moderator-side and participant-side biases with real-world examples of each.
3. What is Confirmation Bias? Explain how it can affect a researcher's interpretation of usability testing results.
4. Explain how the research is integrated into the design process of Empathize, Define, Ideate, Prototype, test and Iterate?
5. Explain Christian Rohrer's three-dimensional framework for categorizing UX research methods (Attitudinal vs. Behavioral, Qualitative vs. Quantitative, Context of Use).
6. What is Bias in User Research? List and explain 4 types of bias that researchers and participants commonly encounter.
7. How does a User experience researcher translate research data into actionable insights?